No of Pages : 3 **Course Code : 23MX13**

Roll No:
(To be filled in by the candidate)

PSG COLLEGE OF TECHNOLOGY, COIMBATORE - 641 004

SEMESTER EXAMINATIONS, MARCH 2025

MCA Semester :1

23MX13 DATA STRUCTURES

Time : 3 Hours **Maximum Marks :100**

INSTRUCTIONS:
1. Answer **ALL** questions. Each question carries 25 Marks.

1. a) i) Differentiate Linear and Non- Linear Data Structures with an example for each. (2)

ii) What is the need for ADT? Define Stack ADT. (3)

b) i) What are the properties of an algorithm? Write an iterative algorithm to print the first 10 Fibonacci numbers. Show that your algorithm possesses all the properties of the algorithm. (5)

ii) Applying Master theorem, find the solution of the following recurrence.

$T(n) = 8T(n/2) + n^3$ (3)

c) i) Define Big O and explain its significance to algorithmic computational complexity. Indicate the complexity of the following code segments using Big O notation. (6)

i)
```
for (i = 0; i<n; i++)
if (i > j)
        sum = sum+1;
else {
        for(k=0;k < n;k++)
            sum=sum-1;
    }
```

ii)
```
void fun(int n) {
        for(int i=0; i *i<n;i++)
        printf ("*");
}
```

iii)
```
for( k =1; k ≤ n; k = 2 * k)
    for( j =1; j ≤ n;j = 2 * j) {
        printf("#");
        break }
```

ii) Solve the following recurrence relation using substitution method and write the complexity using Big O notation. (6)

$T(n)=2T(n-1)+c$

$T(1)=1$

2. a) Explain how the following matrix will be stored using i) row major order and ii) column major order representation. (5)

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 4 & 3 & 2 & 1 \end{bmatrix}$$

Given the base address 1000, size of the integer 2 bytes, find the address of the element 6 in both the representations.

b) Two stacks are to be stored in a single array, as shown in the Fig 1. First stack grows from left and the second from right. Write algorithms to perform basic operations on both stacks. Indicate the situation when the stacks overflow and underflow. (8)

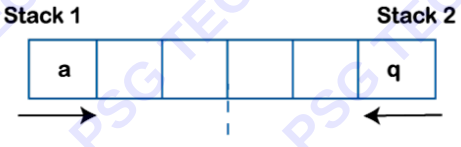

Fig 1

c) i) 1) Write an algorithm insert_SortedList(L,x) to insert an element , x, into a sorted singly linked list L. (6)

2) Write an algorithm insert_CLL(CL, x) to insert element x at the end of the circular linked list CL. (6)

(OR)

ii) 1) Write an algorithm to check the well formedness of parenthesis. Trace with an example. (6)

2) Highlight the difference between linear and circular queue. Write algorithms for the primitive operations on a circular queue. (6)

3. a) What are the different ways of representing a graph? Give an example for the following types of graphs and represent them using each of the representations. (5)

i) Directed graph.

ii) Undirected weighted graph.

b) i) Draw a binary tree whose inorder traversal is B A D C G F E and pre-order traversal sequence is A B C D E F G. Write the sequence in which nodes are traversed in various traversal methods. (4)

ii) Write an algorithm to find the number of leaf nodes in a binary tree. (4)

c) i) What is a binary search tree (BST)? Create a BST by inserting the following elements starting from an initially empty tree: 34, 25, 78, 56, 90, 12. Delete 25 and 56and show the resultant tree. Write an algorithm to insert and delete an element in Binary search tree. What is the best and worst case complexity of searching in a BST? (12)

(OR)

ii) Write algorithms to perform Breadth First and Depth First Search on Graphs. Trace the algorithms on the graph shown in Fig 2.

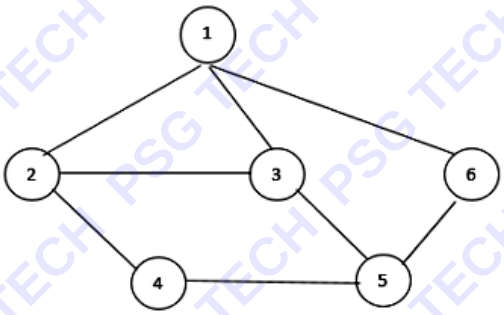

Fig. 2

4. a) Differentiate stable and unstable sort. Is Radix sort stable? Justify your answer with an example. Illustrate Radix sort with the following numbers: 52, 15, 26, 25, 37, 69, 88, 6, 2 (5)

   b) Write algorithms for selection and insertion sort methods. Apply these algorithms to sort the list of elements 12, 25, 3, 45, 56, 34, 70, 27. Show the results of intermediate passes. Write a note on the time complexity of these algorithms. (8)

   c) Describe the term hashing. Given the values {19, 26,13, 48, 17, 3, 14}, a hash table of size 11, and hash function h(x) = x mod 11, show the contents of the hash table using (a) separate chaining (b) linear probing (c) quadratic probing to resolve collisions. Describe the steps involved in insertion, search and deletion in each of these cases. (12)

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FD/RL